

COMMENTS: EM-BASED LIKELIHOOD INFERENCE FOR SOME LIFETIME DISTRIBUTIONS BASED ON LEFT TRUNCATED AND RIGHT CENSORED DATA AND ASSOCIATED MODEL DISCRIMINATION

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We would like to take this opportunity to congratulate Professors Balakrishnan and Mitra on a comprehensive review on the application of the expectation-maximization (EM) algorithm to left-truncated and right-censored data. We also thank the editor, Professor Maxim Finkelstein, for inviting us to discuss this interesting work. We will focus our discussion on using Stochastic Expectation-Maximization (SEM) algorithm, an alternative of EM-algorithm to approximate the maximum likelihood estimates (MLEs), for the problem of interest in the discussion paper.

The SEM Algorithm

The expectation-maximization (EM) algorithm is an iterative procedure that repeatedly fills the missing data in the complete-data log-likelihood with their conditional expected values (E-step) and maximize the complete data log-likelihood to update the parameter estimates (M-step). As illustrated by Balakrishnan and Mitra, the EM-algorithm is efficient in finding the MLEs, especially when computation of the conditional expectation and the maximization are easy to perform. However, there are situations that the conditional expectations in the E-step are intractable or computationally inefficient. For instance, when the underlying distribution is assumed to be generalized gamma distribution, the conditional expectations required for the EM-algorithm may involved integrals or terms with infinite sum (see, Eq. (7.4)) that are not straightforward to evaluate.

Instead of executing the E-step in the EM algorithm, the SEM algorithm proposed by Celeux and Diebolt (1985) completes the observed sample by replacing each missing datum by a value randomly drawn from the distribution conditional on results from the previous step. The SEM algorithm is appealing when the conditional expectation of the missing data is intractable while the censored data is easy to impute. Based on results from both theoretical work and real applications, estimators from the SEM algorithm is almost identical to the maximum likelihood estimates (Nielsen, 2000; Ye

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and Ng, 2014; Zhang, Ye and Xie, 2014). Nevertheless, because of the stochastic nature, the SEM algorithm will not be trapped in a saddle point, which is a serious problem for the EM algorithm.

Implementation

Here, we illustrate the implementation of SEM for left-truncated and right-censored data when the underlying distribution is generalized gamma distribution. Following the notation and the motive example used in Balakrishnan and Mitra, let $\mathbf{A}_1$ and $\mathbf{B}_1$ be the sets of observed and right-censored data in the set of machines installed after 1980 ($\mathbf{S}_1$) without left truncation, and $\mathbf{A}_2$ and $\mathbf{B}_2$ be the sets of observed and right-censored data in the set of machines installed before 1980 ($\mathbf{S}_2$) with left truncation, respectively. We further denote the cardinalities of the sets $\mathbf{A}_1$, $\mathbf{B}_1$, $\mathbf{S}_1$, $\mathbf{A}_2$, $\mathbf{B}_2$ and $\mathbf{S}_2$ as

$$\begin{aligned} |\mathbf{S}_1| &= \sum_{i=1}^n v_i = n_1, |\mathbf{A}_1| = \sum_{i=1}^n v_i \delta_i = r_1, |\mathbf{B}_1| = \sum_{i=1}^n v_i (1 - \delta_i) = n_1 - r_1, \\ |\mathbf{S}_2| &= \sum_{i=1}^n (1 - v_i) = n_2, |\mathbf{A}_2| = \sum_{i=1}^n (1 - v_i)(1 - \delta_i) = r_2, |\mathbf{B}_2| = \sum_{i=1}^n (1 - v_i)(1 - \delta_i) = n_2 - r_2. \end{aligned}$$

Given the parameter values $\Theta^{(k)}$ of $\Theta = (\alpha, \eta, \kappa)$ from the k -th SEM cycle, the $(k+1)$ -th cycle for the estimation of parameters of generalized gamma distribution described in Section 7 progresses as follows.

S-step Draw a random sample of size $n_1 - r_1$, $\mathbf{B}_1^{(k)} = \{v_i^{(k)}; i = 1, 2, \dots, n_1 - r_1\}$, from the conditional distribution in Eq. (7.2), and a random sample of size $n_2 - r_2$, $\mathbf{B}_2^{(k)} = \{w_i^{(k)}; i = 1, 2, \dots, n_2 - r_2\}$, then update the pseudo Q -function

$$\begin{aligned} Q(\Theta; \mathbf{A}_1, \mathbf{B}_1^{(k)}, \mathbf{A}_2, \mathbf{B}_2^{(k)}) &= \sum_{i \in \mathbf{A}_1} \ln f_X(t_i) + \sum_{i \in \mathbf{B}_1^{(k)}} \ln f_X(v_i) \\ &\quad + \sum_{i \in \mathbf{A}_2} \ln \left[\frac{f_X(t_i)}{S_X(\tau_i)} \right] + \sum_{i \in \mathbf{B}_2^{(k)}} \ln \left[\frac{f_X(w_i)}{S_X(\tau_i)} \right]. \end{aligned}$$

M-step Maximize the resulting pseudo Q -function, which is a complete data log-likelihood, to obtain $\Theta^{(k+1)}$ for the next cycle.

The S-step imputes a random realization of the missing data conditional on the parameter estimates from the last step, and is easy to implement. The M-step deals with a sample with left truncation only. Existing results on maximization of such a sample can be utilized. Alternatively, we can use direct optimization with the help of statistical software (for example, the function `constrOptim` in R (R Core Team, 2014)). When k is large enough, $\Theta^{(k)}$ is a random variable whose mean is an efficient estimator of Θ (Nielsen, 2000). Therefore, we can obtain $\Theta^{(k)}$, $k = 1, 2, \dots, M$, discard the first B iterations for burn-in and then average over the remaining iterations to get $\hat{\Theta}$. It is easy to verify that for the generalized gamma distribution, the conditions imposed by Nielsen (2000) are satisfied.

Illustration

Based on the data presented in Table A1 in the Appendix, we implement the SEM algorithm to estimate the parameters of generalized gamma distribution. The initial estimates suggested in Section 9.4, $(\alpha^{(1)}, \eta^{(1)}, \kappa^{(1)}) = (14.4, 3.0, 5.4)$, is used. With a burn-in period of 100 cycles and an additional 1000 iterations, we obtained the estimate of (α, η, κ) as $(23.89, 6.51, 1.61)$ with log-likelihood -296.48, compared to the MLE obtained by direct maximization method or EM-algorithm $(\hat{\alpha}, \hat{\eta}, \hat{\kappa}) = (23.82, 6.23, 1.62)$ with log-likelihood -296.31. The evolution paths of the parameters are shown in Figure 1. The paths reveal no obvious trend in the simulation.

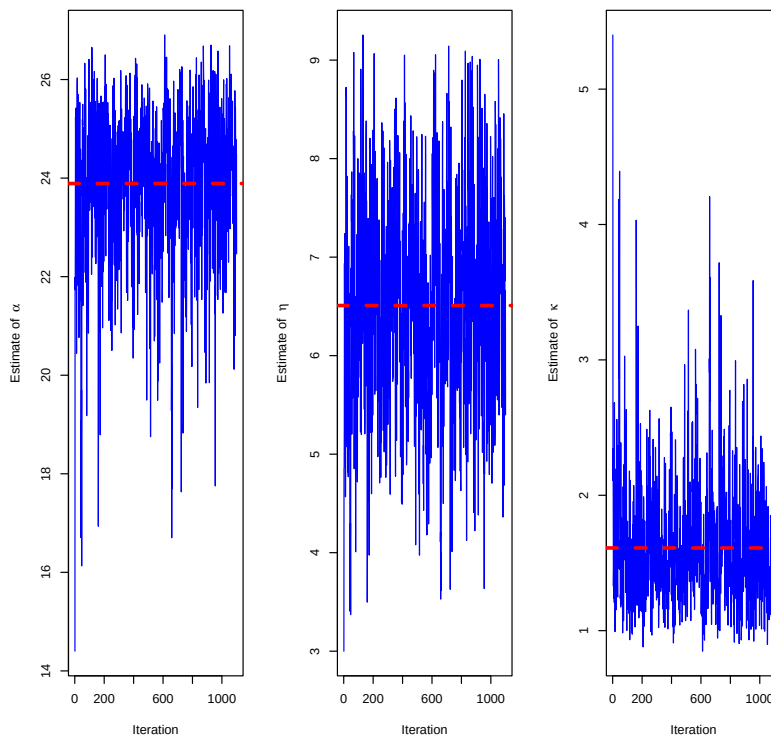


Figure 1: Parameter evolutions in the SEM algorithm for data in Table A1: The dashed-dotted line represents the average of the last 1000 iterations.

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